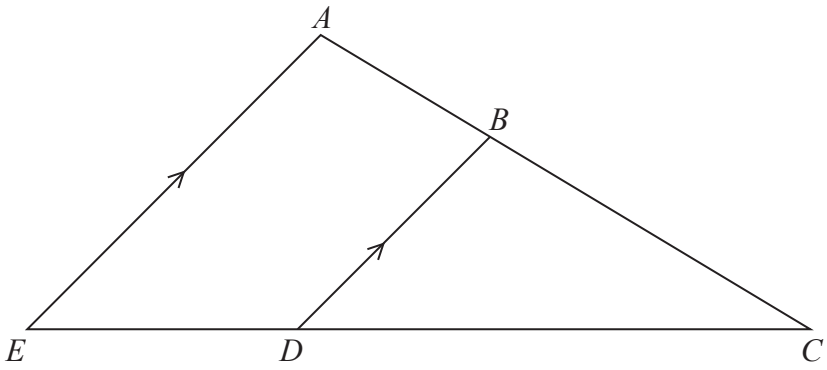
21



ABC and *EDC* are straight lines.
EA is parallel to *DB*.

EC = 8.1 cm.
DC = 5.4 cm.
DB = 2.6 cm.

(a) Work out the length of *AE*.

Similar triangles: DC corresponds to EC; DB to EA.

The scale factor is 8.1 ÷ 5.4 = 1.5.

	Base	Side	
×1.5	5.4	2.6	×1.5
	8.1	3.9	

1 mark
Find a ratio of corresponding sides

1 mark
AE = 2.6 × 1.5 = 3.9 cm

3.9 cm

(2)

AC = 6.15 cm.

(b) Work out the length of *AB*.

BC = AC ÷ 1.5 = 6.15 ÷ 1.5 = 4.1 cm

AB = AC − BC = 6.15 − 4.1 = 2.05 cm

1 mark
Use the same scale factor

1 mark
Subtract BC from AC

2.05 cm

(2)

(Total for Question 21 is 4 marks)